

Bt 504
(Final term Important Questions)

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What is isoelectric point?

- The isoelectric point is the pH at which the net charge of the protein molecule is neutral.
- Different proteins have different isoelectric points.
- Isoelectric point is found by drawing the sample through a stable pH gradient.
- The range of the gradient determines the resolution of the separation.
- This technique is analogous to the first step in 2D-SDS-PAGE.
- In IEF generation of a pH gradient is achieved with soluble ampholytes, which are polycarboxylic acid compounds that generate a stable pH gradient when voltage is applied across the focusing cell.
- The protein sample then is added, voltage again is applied, and the proteins then are separated by isoelectric point.

Two-Dimensional SDS-PAGE

Principle

- It is a single best method for resolving highly complex protein mixtures.
- It involves two different types of separations.
 1. In the first, the proteins are resolved on the basis of isoelectric point by IEF.
 2. In the second, focused proteins then are further resolved by electrophoresis on a polyacrylamide gel.

In past 2D SDS-PAGE was difficult to use because of

- The relative technical difficulty of performing the IEF step.
- Getting the delicate tube gel containing the focused proteins set up to efficiently transfer the proteins in the SDS-PAGE slab gel was a technical challenge.

New designed 2D SDS-PAGE System

- It uses immobilized pH gradient (IPG) strips and relatively foolproof hardware to facilitate the transfer of proteins from the IPG strip into the SDS-PAGE slab gel.
- In this, use of narrow pH ranges facilitates the separation of proteins with highly similar isoelectric points.
- The strip is hydrated with a buffer and the protein is slowly loaded into the strip under voltage.
- Then the voltage is increased to achieve focusing.
- After the focusing step, the strip is treated with a buffer that contains a thiol reductant and SDS and then is joined to the SDS-PAGE slab gel. In this respect, the IPG strip containing the focused proteins acts as a “stacking” gel in 1D-SDS-PAGE.
- The proteins then are resolved on the SDS-PAGE slab gel in the same manner as for 1D-SDS-

PAGE.

- Proteins separated by 2D gels are visualized by conventional staining techniques, including silver, Coomassie, and amido black stains.

Problems

In past 2D SDS-PAGE was difficult to use because of;

- The relative technical difficulty of performing the IEF step.
- Getting the delicate tube gel containing the focused proteins set up to efficiently transfer the proteins in the SDS-PAGE slab gel was a technical challenge.

Gene Regulatory Networks

- Genetic Regulatory Networks (GRNs) are the networks of genes based upon their regulatory interactions with each other
- Generally implemented as Directed Graphs
- Two Genes are connected if the expression of one gene modulates the effect of other gene by either activation or inhibition

Clusters of Orthologous Group (COG)

- Orthologs are assumed to be derived from common ancestor
- Each orthologs might also have paralogs
- Orthologs are clustered to form COG

Greedy Graph Algorithm

Greedy Graph algorithms represent the simplest, most intuitive, solution to the assembly problem.

Steps

Greedy Graph algorithms works as follows:

- Compare all reads or contigs in a pairwise fashion to identify overlapping sequences
- Greedy Graph algorithms works as follows:
- Merge the sequences that overlap each other the best
- Repeat step 2 until no more sequences can be merged, or the remaining overlaps conflict with existing contigs.
- Best overlapping fragments are the one having the highest score
- The scoring function measures the number of matching bases in overlap

Advantages:

- Suitable for small size genomes
- They drastically simplify the graph by considering only the high scoring edges
- May discard each overlap immediately after contig extension

Limitations

- Greedy assemblers can detect false overlaps and high-scoring ones that are resulted from repetitive sequences.

- Graph traversal using greedy approach may cause algorithm to become stuck in local maxima, which produces a suboptimal solution for the assembly problem
- The local maxima will increase the gaps between contigs in the assembly finishing process

Greedy Graph based Assemblers:

- SSAKE
- VCAKE
- SHARCGS

Conclusion:

- Greedy graph approach is simple, easy and fast

ALLPATHS-LG

It's a short read genome assembler from the Computational Research and Development group at the

Broad Institute

Goal

- High quality genome assembly from low cost data
- It can generate high quality genome assemblies using short reads (~100bp) by the new generation of sequencers.

Computational Requirements

- 64 bit linux
- Memory requirements
 - About 160 bytes per genome base implying

Advantages

- Relatively fast runtime
- Can use long reads only for small genomes.

Dicer and Drosha

Dicer

- Dicer or helicase with RNase motif, is an enzyme of RNase III family
- In humans, it is encoded by the DICER1 gene.
- Dicer cleaves double-stranded RNA (dsRNA) and pre-microRNA into short double-stranded RNA

fragments called small interfering RNA and microRNA

- It is able to digest dsRNA into uniformly sized small RNAs (siRNA)
- Dicer family proteins are ATP-dependent nucleases.
- RNase III enzyme acts as a dimer
- Dicer homologs exist in many organisms including C. elegans, Drosophila, yeast and humans
- Amino-terminal helicase domain
- Dual RNase III motifs in the carboxy terminal segment
- dsRNA binding domain

- PAZ domain - 110-130 amino acids

Drosha

- Drosha is a class 2 ribonuclease III enzyme that in humans is encoded by the gene DROSHA (formerly RNASEN) gene.
- The RNase III drosha is the core nuclease that executes the initiation step of microRNA (miRNA) processing in the nucleus.

Dicer and Drosha

- Dicer is an enzyme that is part of the RNase III family
- Drosha belongs to class 2 ribonuclease III enzyme family

Human Dicer :- 1922 aa

Human Drosha:- 1374 aa

Biogenesis of miRNA

- Primary-miRNA is transcribed in the nucleus, and is usually several kilobases long; having 5' cap and a poly-A tail.
- Cleaved in the nucleus by Drosha enzyme to 70nt hairpin transcript (pre-miRNA).
- Transported to the cytoplasm by Exportin 5 through nuclear pores.
- Cleaved by Dicer enzyme (RNase III enzyme) into 19-22nt ds-transcripts.

ORF Finders

Tools are mainly based on pattern finding algorithms

- NCBI's ORF Finder
- ORF Investigator
- OrfPredictor

Protein Structure

• Amino acids

- 20 amino acids
- Hydrophobic / hydrophylic
- Charged / neutral

• Functions

- Enzymes
- Structure protein
- Channel

Types of Proteins

Three major categories

- Fibrous proteins
- Globular proteins
- Complexes with multiple components including proteins

Fibrous Proteins – Cytoskeletal Proteins

- Actin
- Coronin
- Dystrophin
- Keratin
- Tubulin
- Collagen
- Elastin
- Fibronectin

Globular Proteins - Major Types

- Plasma proteins
- Hemoproteins
- Cell adhesion
- Transmembrane transport proteins
- Hormones and growth factors
- Receptors
- DNA-binding protein
- Immune system proteins
- Nutrient storage/transport
- Chaperone proteins
- Enzymes

Complexes with multiple components including proteins

- Nucleosome
- Ribonucleoprotein (generic)
- Signal recognition particle
- Spliceosome

Shotgun sequencing

- The DNA is broken up randomly into numerous small segments, which are sequenced
- Named by analogy with the rapidly expanding, quasi-random firing pattern of a shotgun

Clone by clone Sequencing

Also known as hierarchical shotgun sequencing or map-based shotgun sequencing. The target DNA is first analyzed by clone based physical mapping methods. Individual clones that together span the region of interest are selected and subjected to shotgun sequencing.

1. Pieces of DNA fragments are cloned by using a suitable vector (YAC/BAC)
2. Clones are analyzed for the presence of unique DNA landmarks
3. Minimally overlapping clones are then selected for shotgun sequencing
4. Purified BAC/YAC DNA
5. Physically fragment the DNA
6. Subclone random fragments
7. Generates reads from random subclones

Whole Genome Shotgun Sequencing (WGS):

Whole genome is broken down, cloned into vectors and then sequenced.

1. Physical fragmentation
2. 500bp-1000bp random fragments
3. Subcloned into suitable plasmid vector
4. Sequence reads are generated

Greedy Graph based Assemblers:

- SSAKE
- VCAKE
- SHARCGS

OLC Based Assemblers

- Arachne
- Celera Assembler (CABOG)
- Newbler
- Minimus
- Edena
- CAP

DBG Based Assemblers

- EULER(-SR)
- Velvet
- ALLPATH(S2)
- ABySS
- SOAPdenovo

Difference between Prokaryotic and Eukaryotic gene Expression.

- mRNA is not modified
- The existence of introns in prokaryotes is extremely rare.
- In prokaryotes, the newly synthesized mRNA is polycistronic (code for more than one polypeptide chain).
- In prokaryotes, transcription of a gene and translation of the resulting mRNA occur simultaneously.
- Many polysomes are found associated with an active gene.

Prokaryotes-operons simultaneously transcription and translation

- 27% of E. coli genes (Housekeeping genes not in operons)

Eukaryotes – no operons, RNA processing, chromatin remodeling.

- More kinds of control elements
- RNA processing
- Chromatin remodeling
- Histones must be modified to loosen DNA

Genome and Genomics

The complete set of DNA found in each cell is known as the genome and study is called as genomics.

Proteome and Proteomics

The complete set of proteins found in each cell is known as the proteome.

Proteins concentration (and activity) may be different than gene expression due to post-translational modification.

ORF significance

- Helps finding orthologs, Gene families and domains
- Proteins with highly significant alignment score are likely to be orthologs
- Mostly the proteins related to core biological functions

Converging and Diverging Paths

- Inverse definition than for the bubbles, two paths converge into one, that later diverges again into two separate paths
- Possible causes are repeats in the target genome

Spurs

- Short dead-end branches (divergences) of the main path
- Possible causes include sequencing errors toward one end of a read, and low coverage

Bubbles

- Divergence of a path into two branches that afterwards join together again into one path
- Possible causes include sequencing errors toward the middle of a read, and by polymorphisms in the target.

OLC Advantages

- Suitable for large size genomes
 - Works for both short and long reads
 - OLC approach is capable of handling NGS data
 - Generate correct order of contigs
 - OLC recognizes repeats but may collapse them into one
- ABBBCBBD => ABCBD

Advantages of single molecule sequencing

- Less sample preparation (no PCR)
- No amplification
- no PCR errors

- fewer contamination issues
- no GC-bias
- analyze every sample (unPCRable / unclonable)
- analyze low quality DNA (museum, archeological, forensics samples)
- Absolute quantification
- Sequence RNA directly

TSS Prediction

Translation Start Site (TSS)

Translation starts with ATG that codes for methionine in a polypeptide

Assumption

Certain nucleotides prefer to be around TSS than others.

The “biased” nucleotide distribution information is a basis for translation start prediction

Coding Potential

Hexamer frequencies in coding versus non-coding regions may provide important insights

Frequency of X(A,G,C,T) at position i is

$$F_i(X) = \log(C_i(X)/N_i(X))$$

Conclusions

- TSS prediction can be an important step in gene prediction
- TSS can be predicted while using the frequency of neighboring nucleotides.

Graph Algorithms for Assembly

- The de Bruijn Graph (DBG) methods use some form of K-mer graph
- The greedy graph algorithms may use OLC or DBG.

NGS assemblers have been organized into three categories, based on graphs:

- Overlap-Layout-Consensus (OLC) methods rely on an overlap graph

Orthologs

Genes in two organisms that are so similar that they must have the same function and evolutionary history are called orthologs (Fitch, 1970).

Paralogs

Gene families originating from rare gene duplication events over the evolutionary time are called paralogs.

Unlike pseudogenes, 2nd copy of gene remains functional.

Genome Network

Directed Graphs

- If the edges may only be traversed in one direction, the graph is known as directed graph

Undirected Graphs/Bidirected graphs

- Undirected graph is set of objects (called vertices or nodes) that are connected together, where all the edges are bidirectional.

Protein Extraction

Protein is generally extracted with least contamination with the aid of

1. Detergents
2. Reductants
3. Denaturing agents
4. Enzymes

Detergents

- Detergents e.g SDS, 3- ([3-cholamidopropyl] dimethylammonio)-1propane sulfonate (CHAPS), cholate, Tween can be used for protein extraction.
- These help to solubilize membrane proteins and aid their separation from lipids

Reductants

- Reductants e.g. e.g., dithiothreitol [DTT], mercaptoethanol, thiourea are used.
- These reduce disulfide bonds or prevent protein oxidation

Denaturing agents

- Examples of denaturing agents are urea and acids etc.
- These agents disrupt protein protein interactions, secondary and tertiary structures by altering solution ionic strength and pH

Enzymes

- Enzymes like DNase and RNase are used.
- Enzymes digest contaminating nucleic acids, carbohydrates, and lipids present in the soup.

Quaternary Structure of Proteins

- Some proteins contain two or more separate polypeptide chains or subunits, which may be identical or different.
- The spatial arrangement of these subunits is known as a protein's quaternary structure.
- A multi-subunit protein is also referred to as a multimer.
- A multimer with just a few subunits is called as oligomer and a single subunit or a group of subunits, is called a protomer.
- Identical subunits of multimeric proteins are generally arranged in a symmetric patterns.
- Oligomers can have either rotational symmetry or helical symmetry.
- There are several forms of rotational symmetry. The simplest is cyclic symmetry, involving rotation about a single axis.
- A somewhat more complicated rotational symmetry is dihedral symmetry, in which a twofold rotational axis is present.
- More complex rotational symmetries include icosahedral symmetry.
- An icosahedron is a regular 12-cornered polyhedron having 20 triangular faces.
- The other major type of symmetry found in oligomers is helical symmetry.

Protein Profiling

- Determination of the proteins that make up a given proteome
- Proteomes vary by cell type
- Proteomes vary by stage of cell development
- Some proteins abundant, others very rare

Protein Profiling Techniques

- Two-dimensional gel electrophoresis
- Chemical protein sequencing
- Protein sequencing by mass spectrometry

454 Pyrosequencing

Sequencing in 454 machines is performed by pyrosequencing

Principle

Based on the generation of light signal through release of pyrophosphate (PPi) on nucleotide addition

$\text{DNA}_n + \text{dNTP} \rightarrow$

$\text{DNA}_{n+1} + \text{PPi}$

PPi is used to generate ATP from adenosine phosphosulfate (APS)

- $\text{APS} + \text{PPi} \rightarrow \text{ATP}$

ATP and luciferase generate light by conversion of luciferin to oxyluciferin.

- Only one of the four nucleotides will generate a light signal that will be recorded on a pyrogram.

Cluster Analysis

- To sort out the relationships of all related proteins
- Clustering classify the proteins based on some objective criteria e.g.
 - E value cut off
 - Distance in alignment

ORF Finding

Long ORF may be a gene

Expected 64/3 ~ 21 codons before we see a stop codon

Genes are longer than this

We might scan for ORF longer than a threshold

Codon usage and likelihood ratio

An ORF is more 'reliable' if it has 'likely' codons

We can do sliding window calculations to ORF having 'likely' codon usage

An ORF is more 'reliable' if it has 'likely' codons

However average vertebrate exon length (130 nucleotides) is too small for reliable peaks

An improvement may be;

In-Frame hexamer count
i.e frequencies of pairs of consecutive codons.

significance of genie

Extra predicted exons can be eliminated based on evidence from homology searches.

Likelihood scores provided for each predicted exon.

- The model includes information about:
- Average length of exons and introns.
- Compositional information about exons and introns.
- A neural-net derived model of splice junctions and consensus sequences around splice junctions.
- Splice junction information can be further improved by including results of homology searches.

Genie is approximately 60- 75% accurate on eukaryotic genomes.

Biological Importance of proteins

- Cellular Function
- Structural importance
- Enzymes
- Hormones
- Transport protein
- Messenger protein
- Antibodies
- Plasma protein
- Protein in diet

Cellular Function

- Structural support
- Biochemical reactions of cell
- Role of proteins in plasma membrane

Structural Support

- Microfilaments:
- Distribution of Organelles
- Formation of mitotic Spindle
- Actin Filaments : Cell movements
- Intermediate Filaments
- Architectural support inside cells

Example of some Important Hormones

- Testosterone
- Estrogen

- Growth Hormone (GH)
 - Follicle Stimulating Hormone (FSH)
 - Thyroid Hormone (TH)
 - Melanocyte Stimulating Hormone(MSH)
 - Anti Diuretic Hormone (ADH)
 - Prolactin
 - are generally small proteins, many hormones. e.g.
- Oxytocin Occurs females and stimulates uterine

Transport Proteins

- These are often Globular type of proteins
- Generally tightly packed with polar side group on the outside of molecule. e.g.
- Serum albumin
- Myoglobin
- Hemoglobin

Enzymes

There are three types of enzymes

- Metabolic enzymes
- Digestive enzymes
- Food enzymes

Messenger Proteins

- These are generally quiet small proteins many are harmones e.g
- Oxytocin:
- Occurs in female and stimulate uterine contraction
- VESSOPRESIN ;
- major function as anti diuretic
- each of these has 9 amino acids
- Insulin : regulate the glucose metabolism

Antibodies

- These are protein components of immune system.
- Have main function to bind antigens.

Plasma Proteins

- Albumin
- Made mainly in liver.
- Helps to keep the blood from leaking out of blood vessels.
- Help to carry medicines and other substances.
- Important for tissue growth and healing.
- Globulin
- Made up of different proteins i.e. alpha, beta and gamma types.
- Have a role in immunity.
- Determines chances of developing an infection.

Proteins in Diet

Sources of Proteins

- Meat
- Beef
- Poultry
- Fish
- Egg
- Beans
- Milk

Pharmaceutical Importance of proteins

- Proteins as pharmaceuticals
- Proteins applications
- Whey proteins health effects
- Iron chelate Protein
- Zinc chelate Protein
- Tumor markers

Yeast Genome: Genome of Yeast Cell

The largest known regulatory sequences are spread over about 2,800bp

MUC1/FLO11

Yeast genes have names consisting of three letters and up to three numbers

GPD1, HSP12, PDC6

Usually they are meaningful

Yeast Genome – Genes Nomenclature

Wild type genes are written with capital letters in italics: *TPS1*, *RHO1*, *CDC28*

Recessive mutant genes are written with small letters in italics: *tps1*, *rho1*, *cdc28*

Three letters provides information about a function, mutant phenotype, or process related to that gene.

CDC - Cell Division Cycle ; ADE-ADENine biosynthesis

Important Hormones

- Testosterone
- Estrogen
- Growth Hormone (GH)
- Follicle Stimulating Hormone (FSH)
- Thyroid Hormone (TH)
- Melanocyte Stimulating Hormone(MSH)
- Anti Diuretic Hormone (ADH)
- Prolactin
- generally small proteins, many hormones. e.g. Oxytocin Occurs females and stimulates uterine.

Limitations of Restriction Mapping

In practice, if a DNA molecule is less than 50 kb in length it is usually possible to construct an restriction map.

To construct map for large genomic size is difficult

The limitations of restriction mapping can be eased slightly by choosing enzymes expected to have

infrequent cut sites(rare cutter) in the target DNA molecule.

Objectives of HGP

Create a genetic and physical map of the 24 human chromosomes (22 autosomes, X & Y)

Identify the entire set of genes & map them all to their chromosomes

Determine the nucleotide sequence of the estimated 3 billion base pairs

Analyze genetic variations in humans

Map and sequence the genomes of model organisms

Types of RNA

– Housekeeping RNAs

Ribosomal RNA

Transfer RNA

Small nuclear RNA

Small nucleolar RNA

– Small Non-coding RNAs

– Long Non-coding RNAs

Translation is the process of translating the sequence of a messenger RNA (mRNA) molecule to a sequence of amino acids during protein synthesis. The genetic code describes the relationship between the sequence of base pairs in a gene and the corresponding amino acid sequence that it encodes.

Translation involves four steps:

Initiation: The DNA molecule unwinds and separates to form a small open complex. RNA polymerase binds to the promoter of the template strand.

Elongation: RNA polymerase moves along the template strand, synthesising an mRNA molecule. In

prokaryotes RNA polymerase is a holoenzyme consisting of a number of subunits, including a sigma factor

(transcription factor) that recognises the promoter. In eukaryotes there are three RNA polymerases: I, II and

III. The process includes proofreading mechanism.

Termination: In prokaryotes there are two ways in which transcription is terminated. In Rho-dependent

termination, a protein factor called "Rho" is responsible for disrupting the complex involving the

template

strand, RNA polymerase and RNA molecule. In Rho-independent termination, a loop forms at the end of the RNA molecule, causing it to detach itself. Termination in eukaryotes is more complicated, involving the addition of additional adenine nucleotides at the 3' of the RNA transcript (a process referred to as polyadenylation).

Processing: After transcription the RNA molecule is processed in a number of ways: introns are removed

and the exons are spliced together to form a mature mRNA molecule consisting of a single protein-coding

sequence. RNA synthesis involves the normal base pairing rules, but the base thymine is replaced with the base uracil.

Transposable Elements (TEs)

Large portion of eukaryotic genome

Thought to play important role in evolution of these genomes

Move (Jump) from one location to another faster than chromosome replicate

Have a potential to increase in number

Make up a large proportion of eukaryotic genome

Detectable until blend into genome due to mutations

Vast majority of TEs can be classified into four families:

LINEs (Long Interspersed Nuclear Elements, autonomous)

SINEs (Short Interspersed Nuclear Elements, use LINE proteins for life cycle)

LTR elements (Long Terminal Repeats; derived from retroviruses)

DNA transposons (replicate without RNA intermediary)

High-Performance Liquid Chromatography

Introduction

- High-performance liquid chromatography (HPLC; formerly referred to as high-pressure liquid chromatography), is a technique in analytical chemistry used to separate, identify, and quantify each component in a mixture.

- HPLC is a chromatographic technique that can separate a mixture of compounds. It is used in biochemistry and analytical chemistry to identify, quantify and purify the individual components of a mixture.

- Although HPLC of intact proteins has not become a widely used technique for analytical proteomics, it is nevertheless highly applicable as an initial step to fractionate protein mixtures.

- HPLC would appear to be about as useful as preparative IEF for resolving protein mixtures into fractions.

Principle

- HPLC relies on pumps to pass a pressurized liquid solvent containing the sample mixture through a column

filled with a solid adsorbent material.

- Each component in the sample interacts slightly differently with the adsorbent material, causing different flow rates for the different components and leading to the separation of the components as they flow out the column

Advantages of HPLC

- The advantage of HPLC is the diversity of separation modes available. Indeed, tandem HPLC separations combine two different types of chromatography. For example, strong cation exchange, followed by RP (Reverse phase), would apply two completely different separation modes.

Genscan Output

- Poly-adenylation site
- Genscan is 80% accurate on human sequences.
- Eukaryotic gene structures can be quite complex.
- The best approaches to gene finding in eukaryotes combine different methods.

Codons in ORF Finding

Long ORF may be a gene

Expected 64/3 ~ 21 codons before we see a stop codon

Genes are longer than this

We might scan for ORF longer than a threshold.

Microarray Requirements

Microarrays combine genomics, silicon chip manufacturing, DNA and Protein chemistry, signal and image

processing, statistics, software skills and traditional molecular biology experiments.

Names of Standard Amino acids

- Glycine
- Alanine
- Serine
- Threonine
- Cysteine
- Valine
- Leucine
- Isoleucine
- Methionine
- Proline
- Phenylalanine

- Tyrosine
- Tryptophan
- Aspartic acid
- Glutamic acid
- Asparagine
- Glutamine
- Histidine
- Lysine
- Arginine

Design of siRNA

- 21-23 nt dsRNA,
- GC% slightly < 50%
- Perfect complimentary to target mRNA
- Targeting 3'UTR works better than 5' UTR

Applications of siRNA

- Determining protein function
- Easier than preparing knockout
- Used for partial knockdowns
- Applications for cancer prevention, infections and developmental defects **siRNA**
- Exogenously delivered
- 21-23mer dsRNA
- Acts through RISC
- Induces homologous target cleavage
- Perfect sequence match
- Results in target degradation

Homologues

Homologues are thus components or characters (such as genes/proteins with similar sequences) that can be

attributed to a common ancestor of the two organisms during evolution.

Homologues can be

Orthologues

Paralogues

Xenologues

Analogues

Orthologues

Orthologues are homologues that have evolved from a common ancestral gene by speciation.

They usually have similar functions.

Paralogues

Paralogues are homologues that are related or produced by duplication within a genome followed by subsequent divergence.

They often have different functions.

Xenologues

Xenologues are homologous that are related by an interspecies (horizontal transfer) of the genetic material for one of the homologues.

The functions of the xenologues are quite often similar.

Analogues

Analogues are non-homologues genes/proteins that have descended convergently from an unrelated ancestor.

Similar function, different sequence or structure.

Plasma Proteins

Albumin

- Made mainly in liver.
- Helps to keep the blood from leaking out of blood vessels.
- Help to carry medicines and other substances.
- Important for tissue growth and healing.

Globulin

- Made up of different proteins i.e. alpha, beta and gamma types.
- Have a role in immunity.
- Determines chances of developing an infection.

Coverage

The average number of times any given base in the genome is sequenced.

It can be derived by dividing the total length of acquired sequences by the genome length

$$C = NL / G$$

N: Total number of reads

L: Length of a Read

G: Genome size

Represented as X

Example

Illumina produced 1 million reads of 100 bp in length for a species whose genome size is 1 MB.

What is

coverage?

$$C = NL / G \quad 1000000 \times 100 / 1000000$$

$$= 100 \times$$

Alpha - Helix

The simplest arrangement which a polypeptide chain could assume with its rigid peptide bonds is a helical structure, which Pauling and Corey called the α -helix.

The helical twist of the α -helix found in all proteins is right-handed.

The repeating unit is a single turn of the helix, which extends about 5.4 Å (includes 3.6 amino acid residues) along the long axis.

The amino acid residues in an helix have conformations with $\psi = -45$ to -50 and $\phi = -60$.

An helix makes optimal use of internal hydrogen bonds.

About one-fourth of all amino acid residues in polypeptides are found in α -helices while in some proteins it is the predominant structure.

Enzymes types

There are three types of enzymes

- Metabolic enzymes
- Digestive enzymes
- Food enzymes

Strategies for performing Assembly

Assembler

- Newbler
- Celera or PBJelly
- ALLPATHS-LG

siRNA

- Exogenously delivered
- 21-23mer dsRNA
- Acts through RISC
- Induces homologous target cleavage
- Perfect sequence match
- Results in target degradation

miRNA

- Endogenously produced
- 21-23mer dsRNA
- Acts through RISC
- Induces homologous target cleavage
- Imperfect sequence match
- Results in translation arrest

Radiation Hybrid Mapping of Genomes

A radiation hybrid is a rodent cell line carrying a small genomic DNA molecule from another organism e.g. a human.

Exposure to X rays breaks the DNA in human cells.

The fragments become smaller with more X ray exposure

Fragments length determines the map resolution

Irradiation kills the human cells, which are then fused with rodent cells, rescuing chromosomal fragments that are typically a few Mb in length.

Human DNA in the RH is analyzed for gene and/or DNA markers.

Closer the two markers to each other – chromosome; the more likely they are to be found together in an RH.

Capillary Electrophoresis (CE)

- Capillary electrophoresis (CE) operates on the same general principal as IEF.
- Proteins placed in an electric field will migrate to a point in a pH gradient where they display an overall neutral charge.
- The performance of the analysis in a microcapillary tube provides greatly enhanced resolution over the preparative IEF techniques discussed earlier.
- CE offers the greatest resolution of all peptide analytical techniques and can be coupled directly to MS instruments.
- CE thus has great potential as a technique for analytical proteomics.
- The utility of CE is limited at the present time by the lack of commercially available, robust, and reliable CE-MS instrumentation for analytical proteomics.
- Development of instrumentation for this purpose is continuing and CE-MS may become a very useful tool in proteomics analysis in the near future.

Working principle

- Capillary tube is placed between two buffer reservoir, and an electric field is applied, separation depends on electrophoretic mobility & electro-osmosis .
- Defined volume of analyte is introduced in to the capillary by replacing one buffer reservoir with sample vial.
- Electrophoretic separation is measured by detector.

Structural Proteomics

- Structural proteomics deals with mapping out the 3-D structure and nature of protein and/or proteins complexes.

Functional Proteomics

- Posttranslational modifications
- Protein-protein, protein-ligand interactions
- Sequence-structure-function relationships

Expression Proteomics

- Expression proteomics refers to the quantitative study of protein expression between samples differing by some variable.

- Analysis of differential protein expression.

Post translation of RNA

Proteins are synthesized by the translation of mRNAs into polypeptides on ribosomes. In most cases, the initial polypeptide-translation product undergoes some type of modification before it assumes its functional role in a living system.

These changes are broadly termed “posttranslational modifications” and encompass a wide variety of

reversible and irreversible chemical reactions.

Approximately 200 different types of posttranslational modifications have been reported.

DNA Based Microarray Chips

Gene expression profiling

Discovery of drugs

Diagnostics and genetic engineering

Alternative splicing detection

Functional genomics

DNA sequencing

Toxicological research (Toxicogenomics)

SMRT sequencing

- Single Molecule Real Time Sequencing
- utilizes the zero-mode waveguide (ZMW), developed in the laboratories of Harold G. Craighead and Watt W. Webb at Cornell University
- ZMW guides light energy into a volume that is small in all dimensions compared to the wavelength of the light
- Nanophotonic structure, a circular hole in an aluminum film on a silica substrate
- With an active polymerase immobilized at the bottom of each ZMW, nucleotides diffuse into the ZMW chamber.
- A, C, G and T are labeled with a different fluorescent dye having a distinct emission spectrum
- Nucleotides held by the polymerase prior to incorporation emit an extended signal that identifies the base being incorporated
- utilizes the zero-mode waveguide (ZMW), developed in the laboratories of Harold G. Craighead and Watt W. Webb at Cornell University

Automated Sequencing

- Florescent dyes are used to label DNA
- Primers for each reaction (ddA, ddT, ddG, ddC) are tagged with separate dyes
- All reactions in same gel lane

- Band colors help in detection
- A fluorimeter and computer are hooked up to the gel and they detect and record the dye attached to the fragments as they come off the gel.
- A fluorimeter and computer are hooked up to the gel and they detect and record the dye attached to the fragments as they come off the gel.

Eulerian Path

- An Eulerian path is a path that visits every edge of the graph once and only once.
- It can end on a vertex different from the one on which it began.

Eulerian Circuit

- An Eulerian path which begins and ends on the same vertex.
- It starts and ends on the same vertex.

Single Gene

Basic physical & functional unit of heredity.

Made up of DNA & contain all genetic information of an organism.

Gene Family

Set of several similar genes, formed by duplication of a single original gene.

Generally have similar biochemical functions.

Cartridge Based

- Spotted
- Electronic
- Spotted glass slide
- Tissue section slide

Types of Genomic Mapping

Genetic Mapping

Approximate location of genes, relative to the other genes

Based on rate of recombination

Physical Mapping

Based on direct analysis of DNA

Places genes on the genomes in relation to distances measured in bp, kbp & mbp.

Gene expression analysis

involves the determination of the pattern of genes expressed at the level of genetic transcription, under specific circumstances or in a specific cell.

Gene expression analysis simultaneously compares the RNA expression levels of multiple genes (profiling) and/or multiple samples (screening). This analysis can help scientists identify the molecular basis of phenotypic differences and to select gene expression targets for in-depth study.

Types of Chromosomes

On the basis of the location of the centromere, chromosomes are classified into four types: metacentric, submetacentric, acrocentric, and telocentric.

1. Metacentric

The centromere is situated in the centre of the chromosome, or median in position. So the two arms are equal in length, and thus chromosome appears V shaped.



2. Sub-metacentric

In this type the centromere is Metacentric sub median in position, or placed near the midpoint. Therefore the arms are unequal in length and resemble the letter 'L' in shape.



3. Telocentric

In this type the centromere is located at one end of the chromosome. And thus such, chromosomes are rod-shaped in appearance.



4. Acrocentric or sub-telocentric

In this type the centromere is located near one end of the chromosome. That is the centromere is sub-terminal in position. The chromosome is rod-shaped.



Structure of Amino acid

Each amino acid has the same fundamental structure, which consists of a central carbon atom, also known as the alpha (α) carbon, bonded to an amino group (NH_2), a carboxyl group (COOH), and to a hydrogen atom.

Why we study genomics?

Genomics helps us understand evolution and protect our biological ecosystems.

Genomics is altering a genome with unparalleled efficiency and precision.

Genomics is fostering an appreciation for what our DNA means for our health, identities and culture.

Gene Assembly

A genome assembly is a computational representation of a genome sequence. Because we are not able to sequence along the complete length of a chromosome, each chromosome assembly is made up of short stretches of sequenced DNA pasted together.

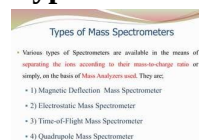
In bioinformatics, sequence assembly refers to aligning and merging fragments from a longer DNA sequence in order to reconstruct the original sequence.

Why we do Genomic Sequencing?

Scientists use a process called genomic sequencing to decipher the genetic material found in an organism or virus. Sequences from specimens can be compared to help scientists track the spread of a virus, how it is changing, and how those changes may affect public health.

Enables us to read a person's individual genome and, among other things, identify differences from the average human genome.

Types of Mass Spectrometry



Applications of Pyrosequencing

Pyrosequencing has been used to correlate DNA methylation with tumor type and gene expression, to measure cellular response to treatment with demethylating agents, and to assess changes in methylation state in relation to tumorigenesis, genetic imprinting, and exposure to environmental toxins.

Pyrosequencing is a unique system that allows the analysis of genetic variations including single-nucleotide polymorphisms, indels and short repeats, as well as assessing RNA allelic imbalance, DNA methylation status, and gene copy number.

Pyrosequencing has potential advantages of accuracy, flexibility, parallel processing, and can be easily automated. Furthermore, the technique avoids the need for labeled primers, labeled nucleotides, and gel electrophoresis.

Mitochondrial genome?

The mitochondrial genome contains 37 genes that encode 13 proteins, 22 tRNAs, and 2 rRNAs. The 13 mitochondrial gene-encoded proteins all instruct cells to produce protein subunits of the enzyme complexes of the oxidative phosphorylation system, which enables mitochondria to act as the powerhouses of our cells.

Mitochondrial DNA (mtDNA) has many special features such as a high copy number in cell, maternal inheritance, and a high mutation rate which have made it attractive to scientists from many fields.

Advantage of Open Reading Frame

One common use of open reading frames (ORFs) is as one piece of evidence to assist in gene prediction. Long ORFs are often used, along with other evidence, to initially identify candidate protein-coding regions or functional RNA-coding regions in a DNA sequence.

Use ORF finder to search newly sequenced DNA for potential protein encoding segments, verify predicted protein using newly developed SMART BLAST or regular BLASTP. This web version of the ORF finder is limited to the subrange of the query sequence up to 50 kb long.

Why helix proteins are stable?

α -helix is right handed and is more stable due to inter molecular H bonding between first and fourth amino acid.

The α helix, a common structural motif of proteins, consists of a right-handed helix with a repeat length of 3.6 amino acid residues per helical turn. The α helix is stabilized by hydrogen bonds between an amide hydrogen of one amino acid and a carbonyl oxygen four amino acids away.

Life cycle of Cell

A cell cycle is a series of events that takes place in a cell as it grows and divides. A cell spends most of its time in what is called interphase, and during this time it grows, replicates its chromosomes, and prepares for cell division. The cell then leaves interphase, undergoes mitosis,

and completes its division.

In eukaryotes, the cell cycle consists of four discrete phases: G1, S, G2, and M. The S or synthesis phase is when DNA replication occurs, and the M or mitosis phase is when the cell actually divides. The other two phases — G1 and G2, the so-called gap phases — are less dramatic but equally important.